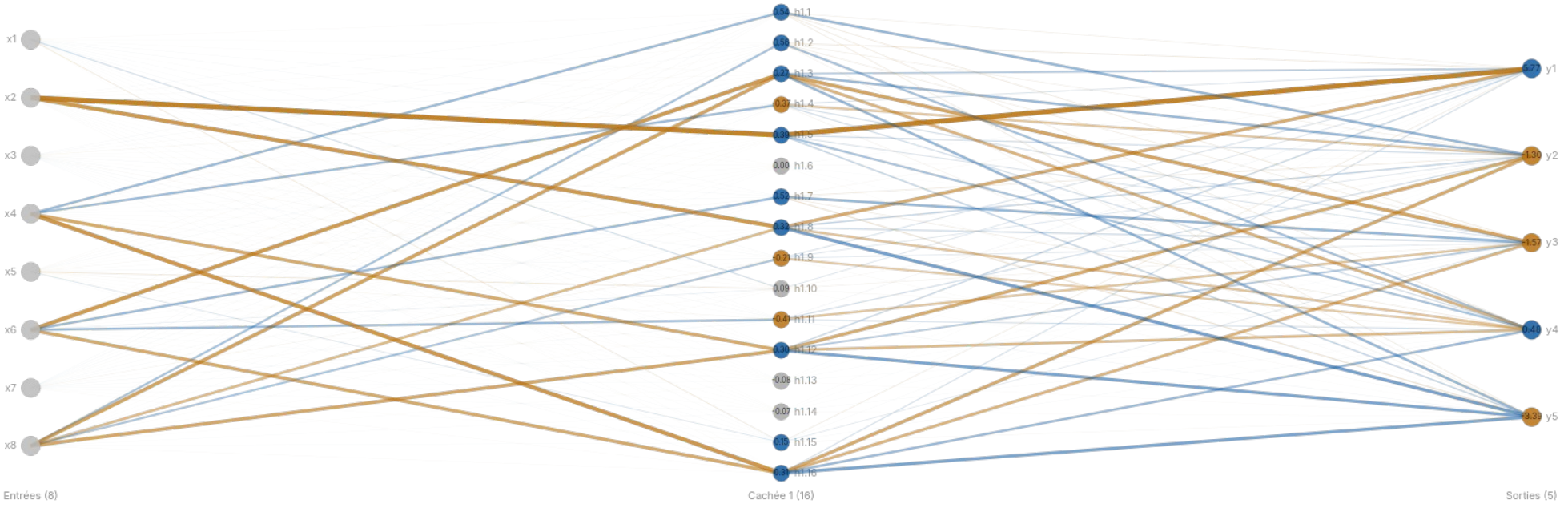
Entrainement d’un réseau de neurones

Le réseau

- 4 utilisateurs => 8 neurones en entrée
- 16 neurones sur la couche intermédiaire
- 5 neurones de fin

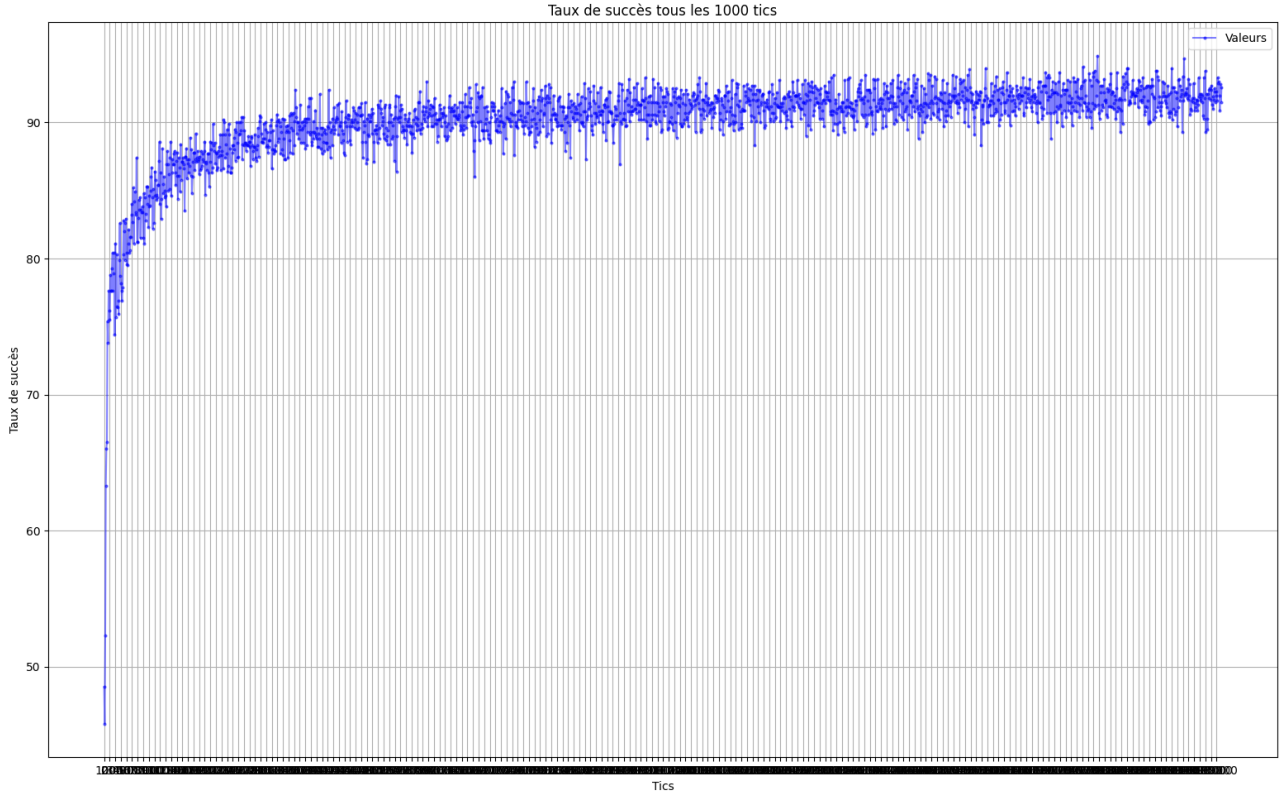


Poids positif Poids négatif Biais > 0 Biais < 0 Biais ≈ 0

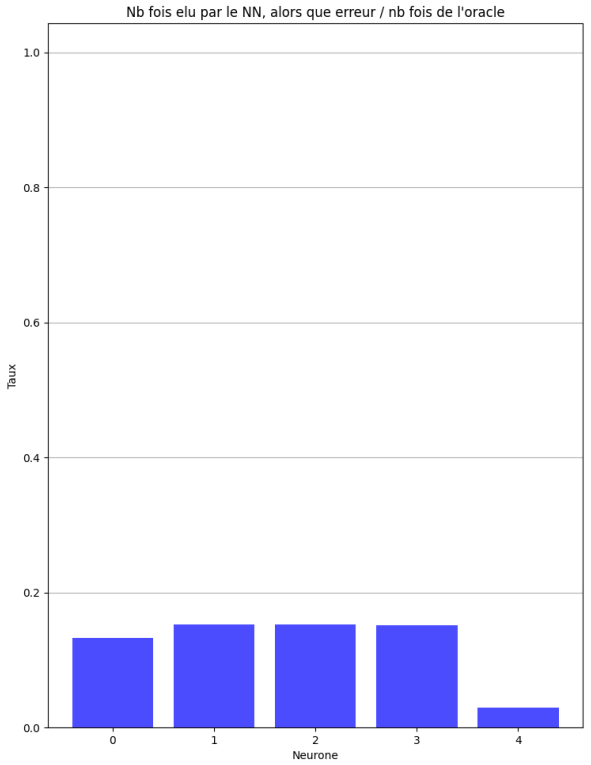
L’entraînement

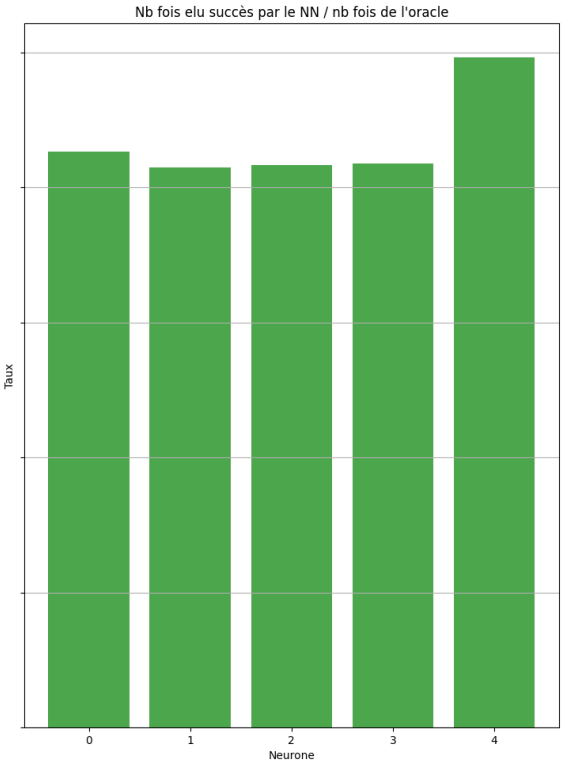
- 2000000 tics
- Tout les 100 tics, ajout de quantités dans les buffers
- À la fin des 200k tics, obtention des résultats et division du learning rate par 2
- Évolution du learning rate : 0.01 → 0.005 → 0.0025 → 0.00125 → 0.0006 → 0.0003 → 0.00015
- Gravité des erreurs : En fonction des inputs et du résultats attendu on classifie la gravité des erreurs ; exemple : $mkn_1 = 15$ et $mkn_2 = 14$, si le réseau de neurones attribue la ressource au user 2 au lieu du 1, comme ils ont un mkn proche c’est pas une erreur grave, la gravité vaut 1. En revanche si les 2 users n’ont rien à transmettre et que le réseau ne donne pas la ressource (neurone de fin n°5) il s’agit là d’un erreur grave don’t la gravité vaut 4.

Taux succès lr = 0.01

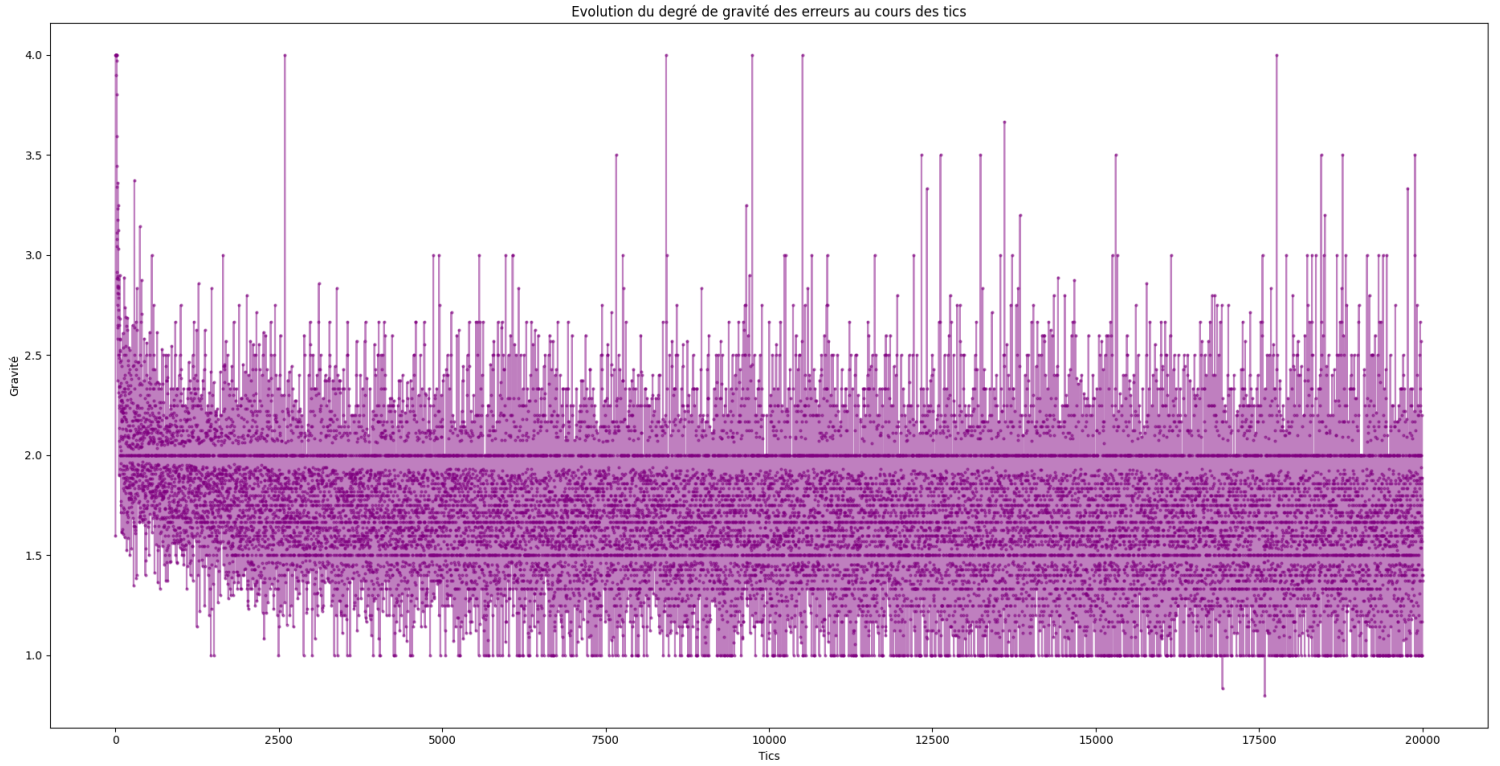


Choix des neurones lr = 0.01

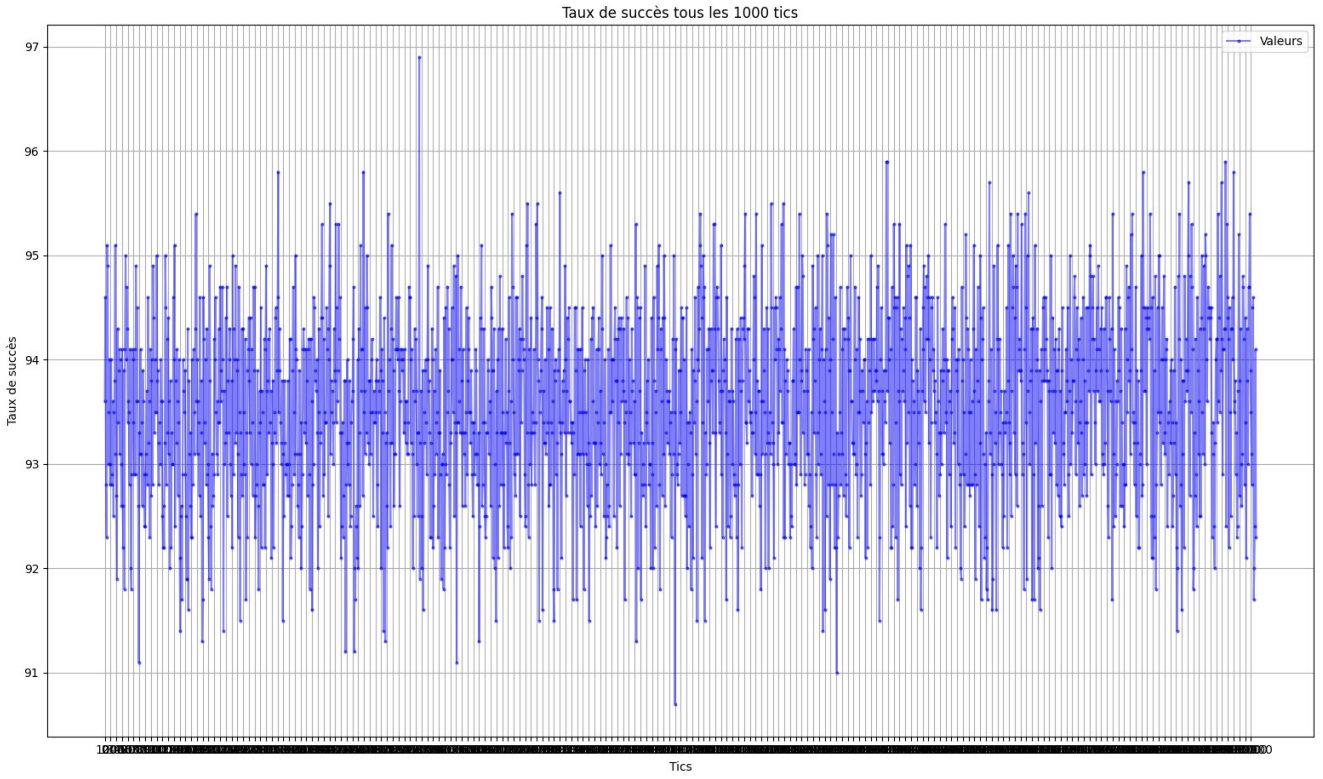




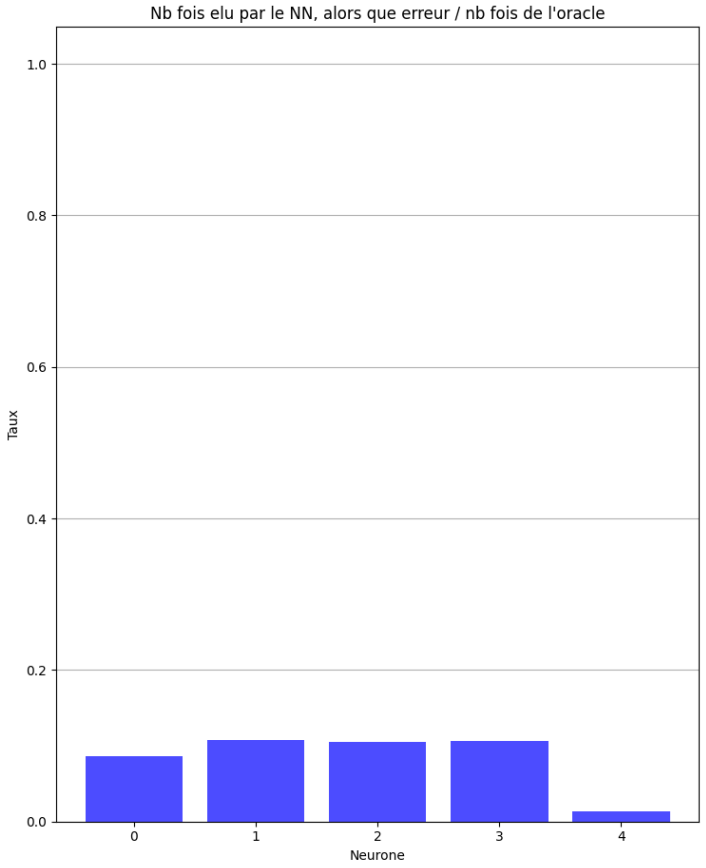
Gravité des erreurs lr = 0.01

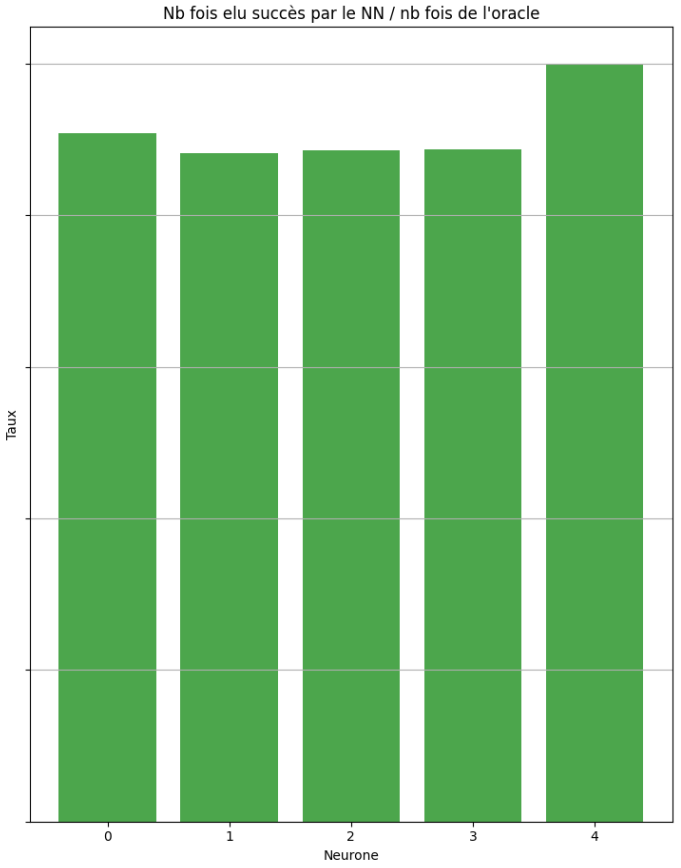


Taux succès lr = 0.005

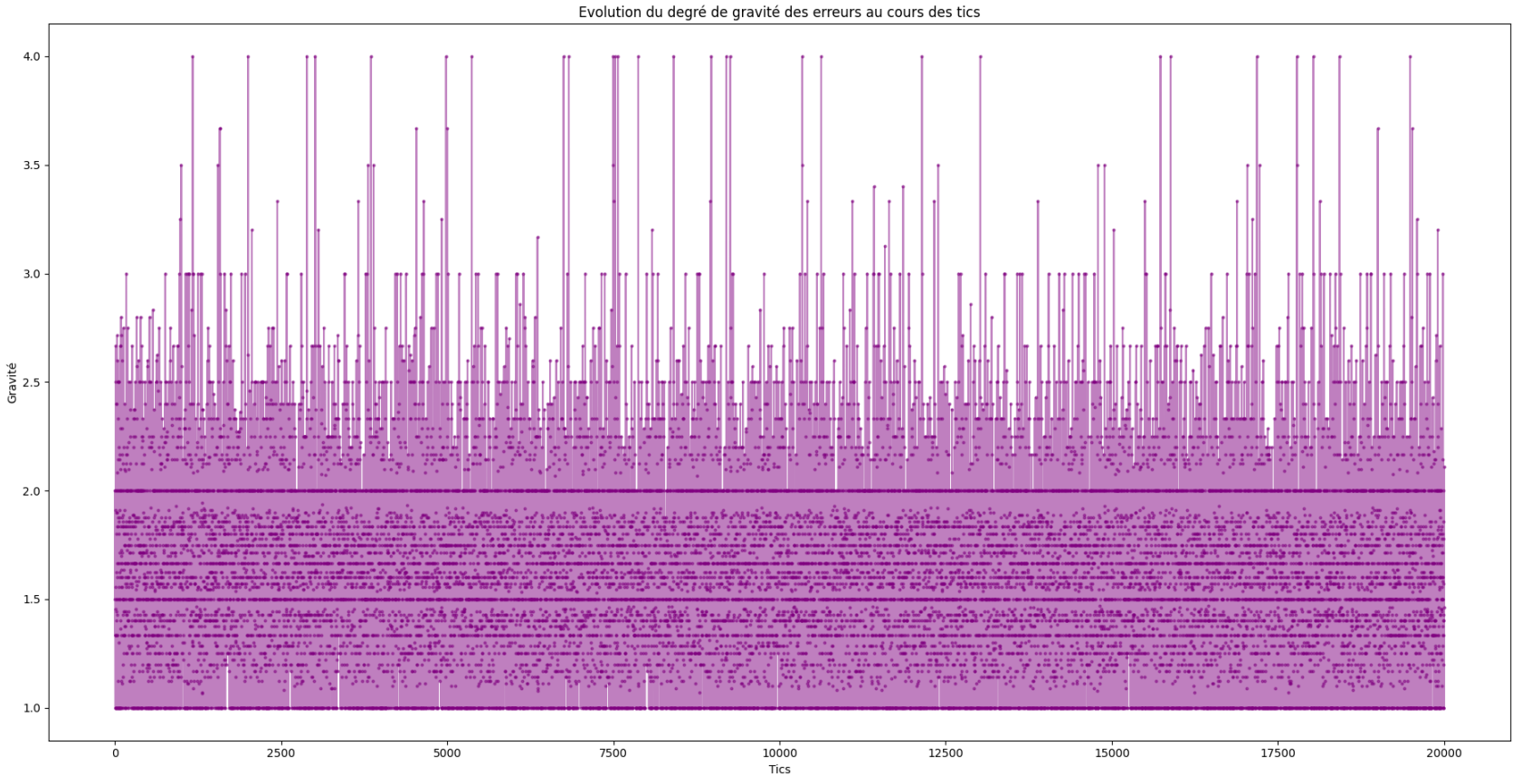


Choix des neurones lr = 0.005

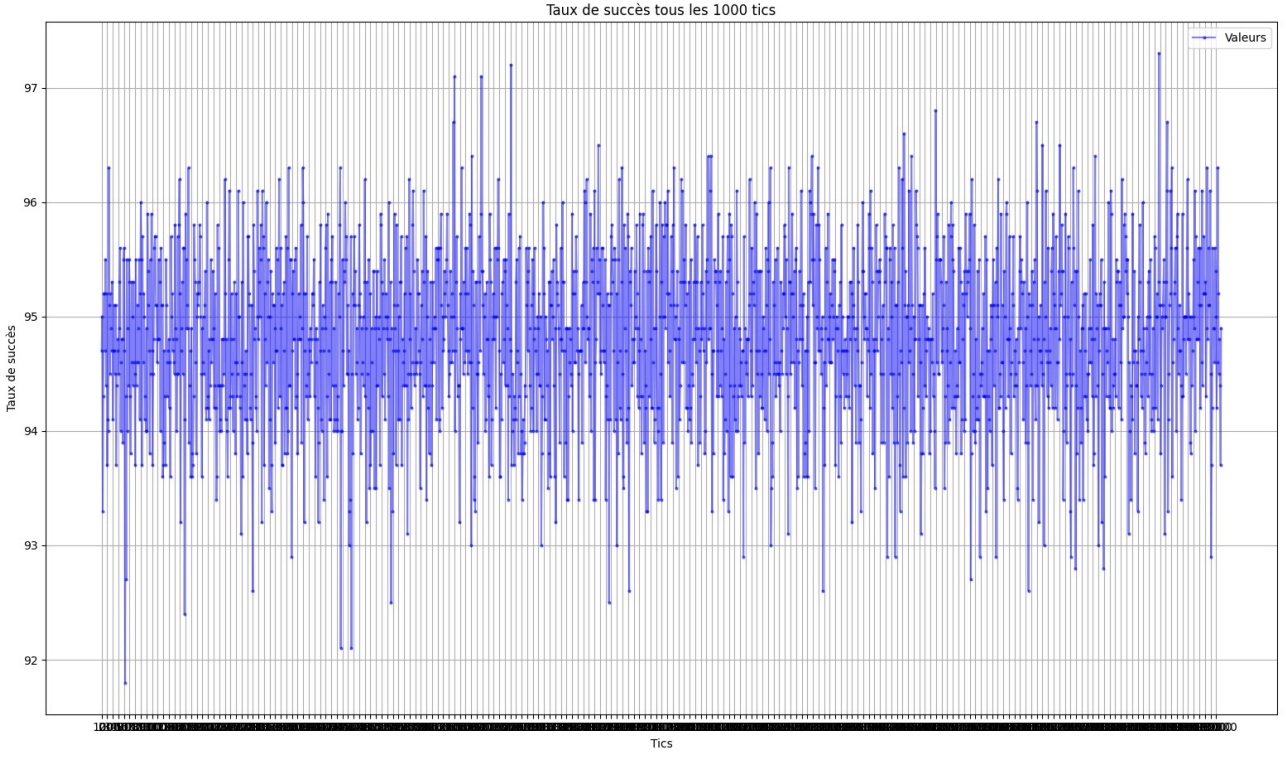




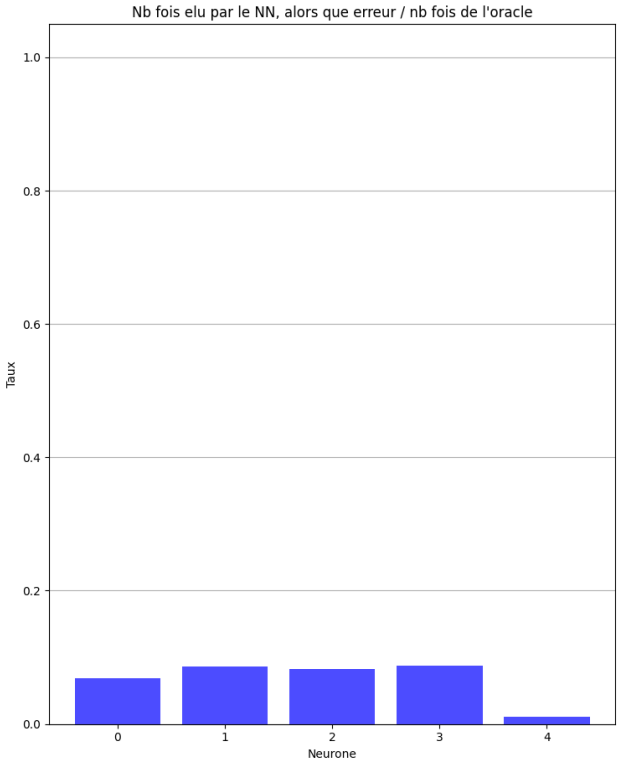
Gravité des erreurs lr = 0.005

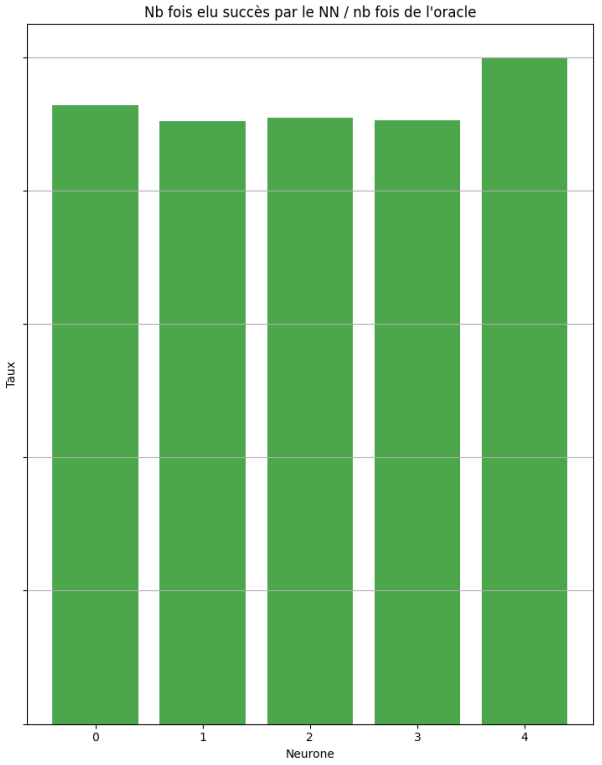


Taux succès lr = 0.0025



Choix des neurones lr = 0.0025

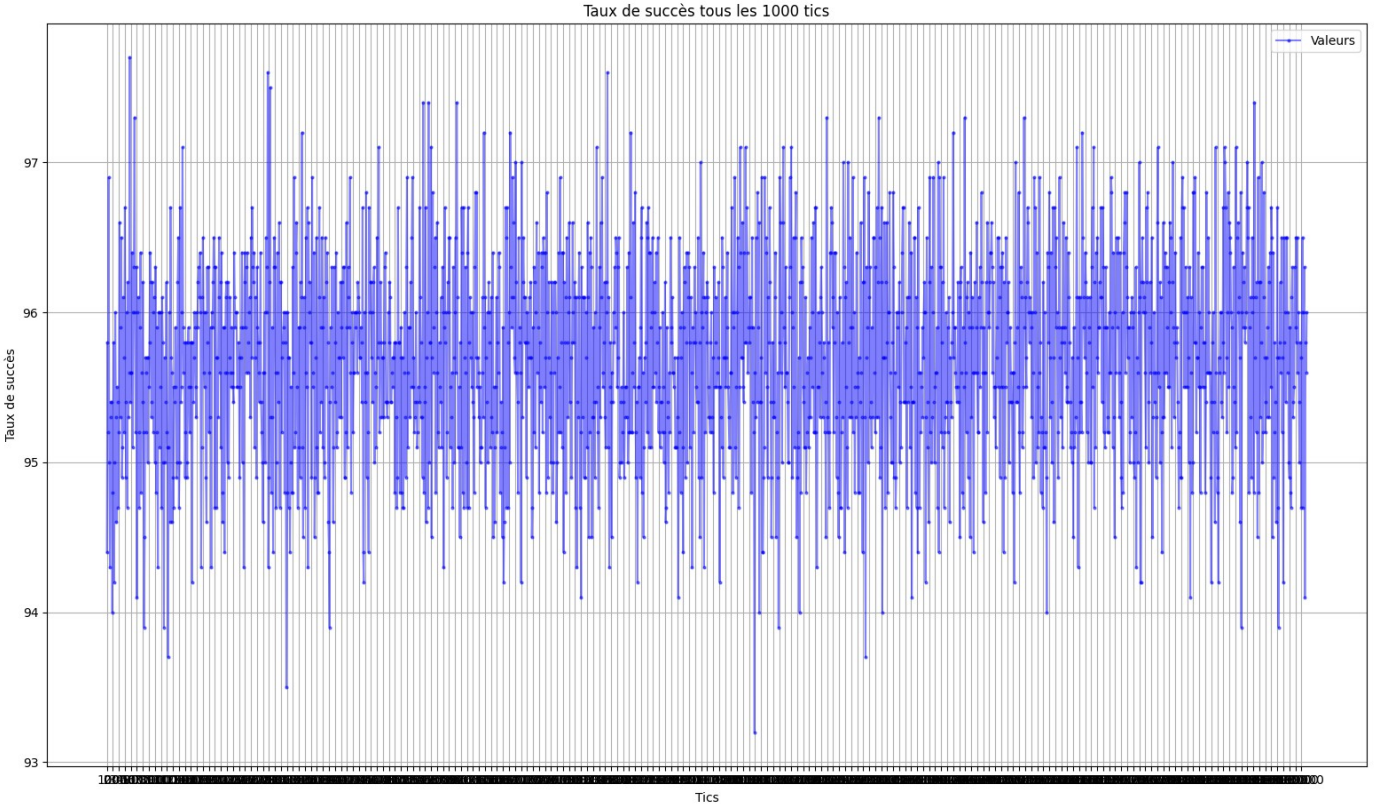




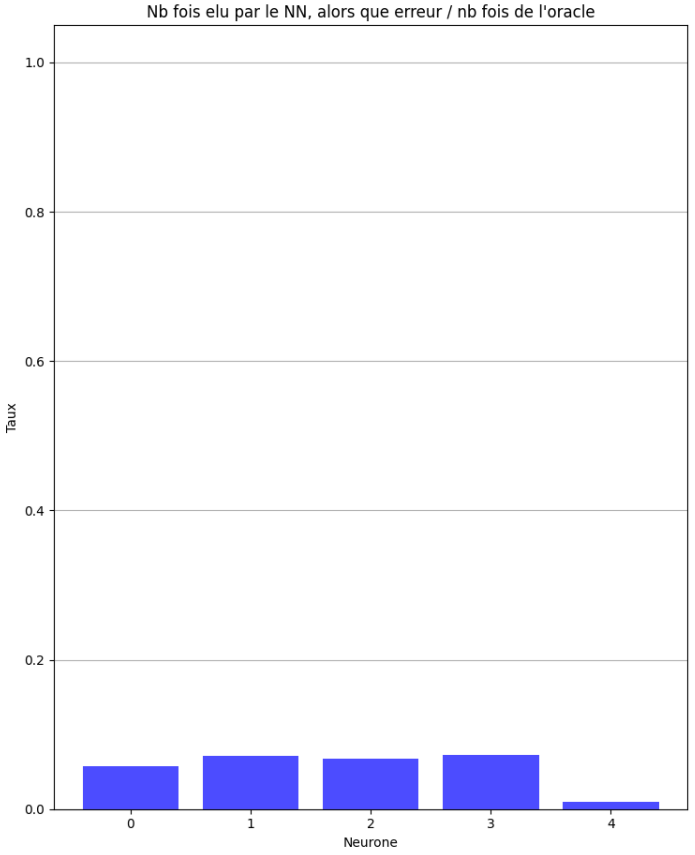
Gravité des erreurs lr = 0.0025

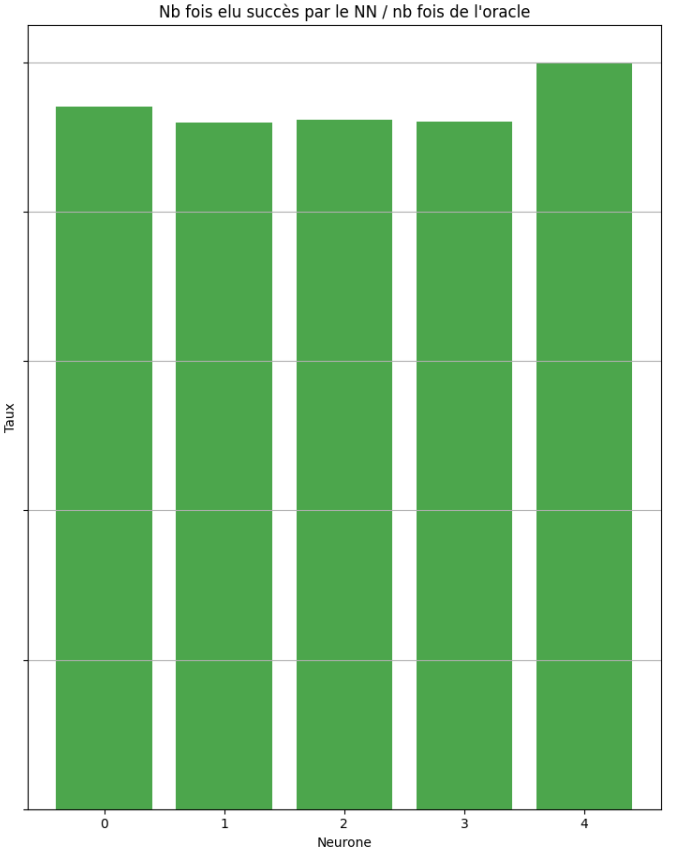


Taux succès lr = 0.00125

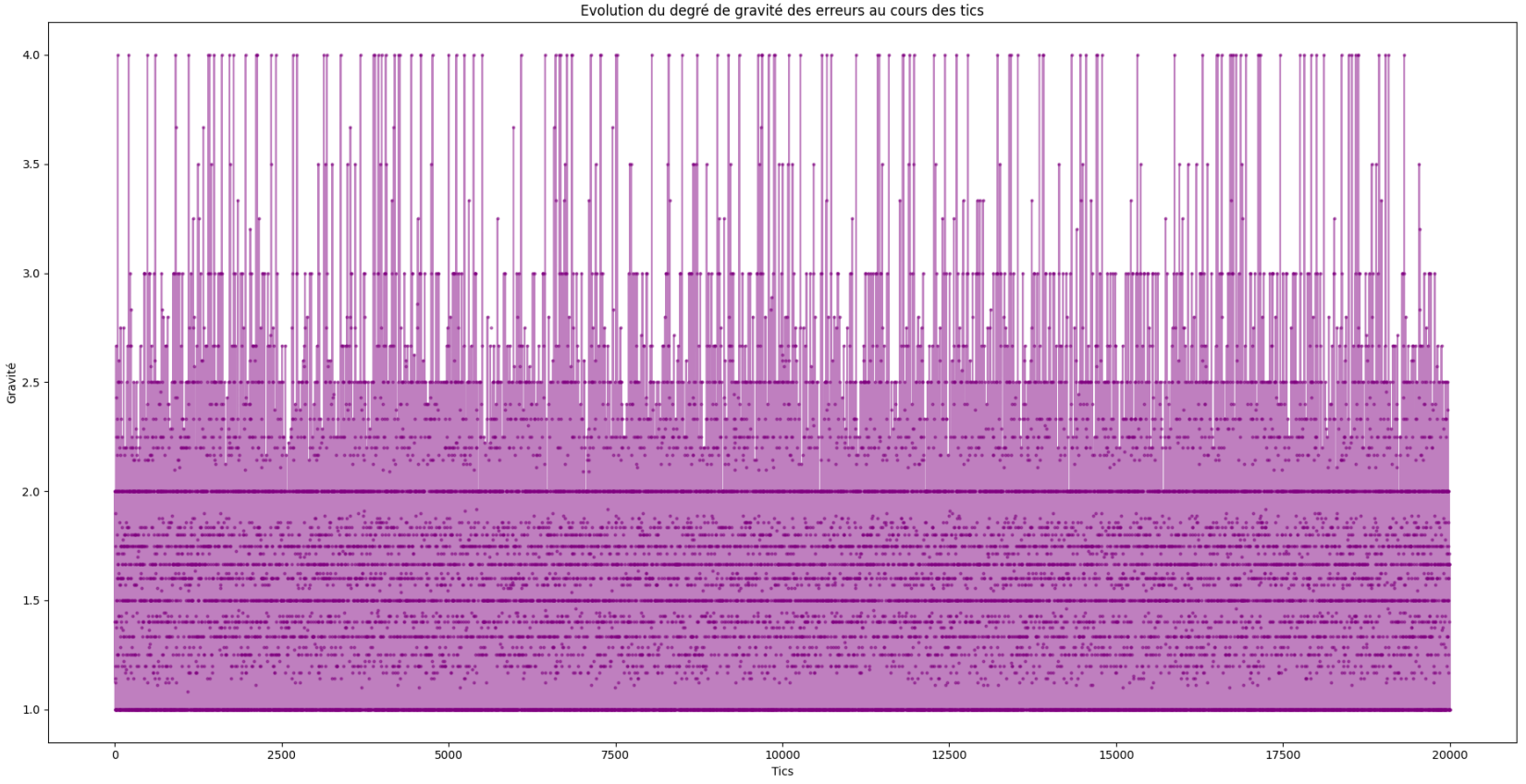


Choix des neurones lr = 0.00125

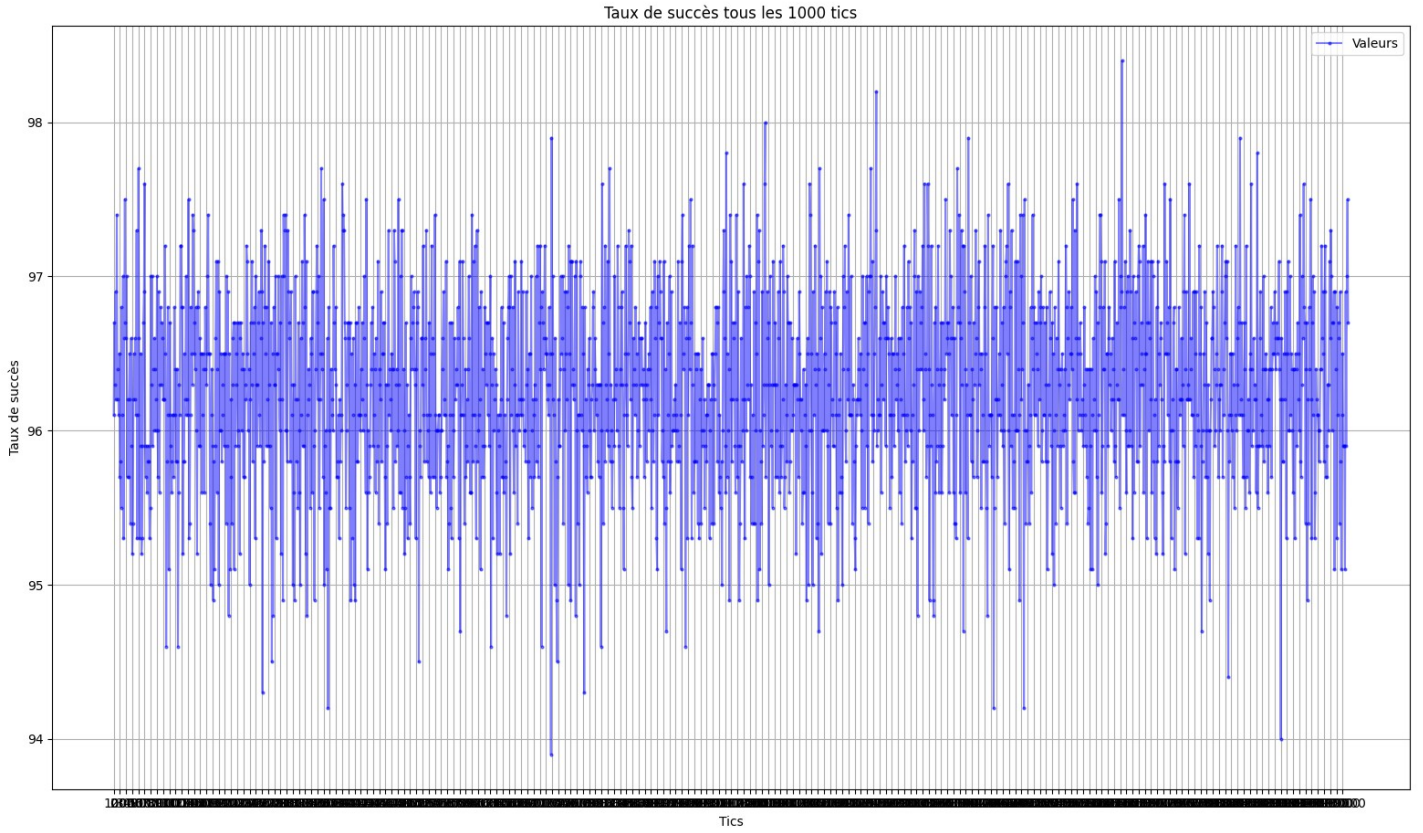




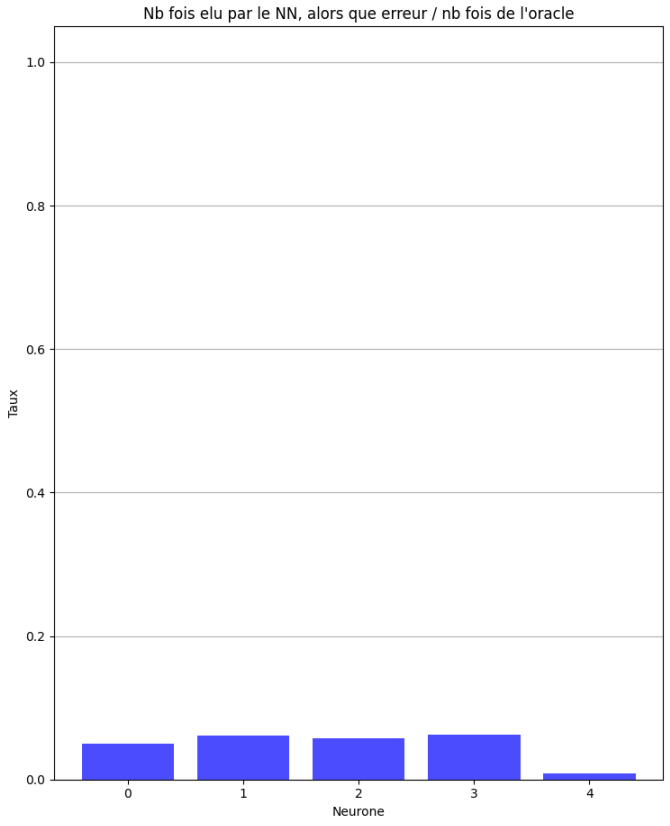
Gravité des erreurs lr = 0.00125

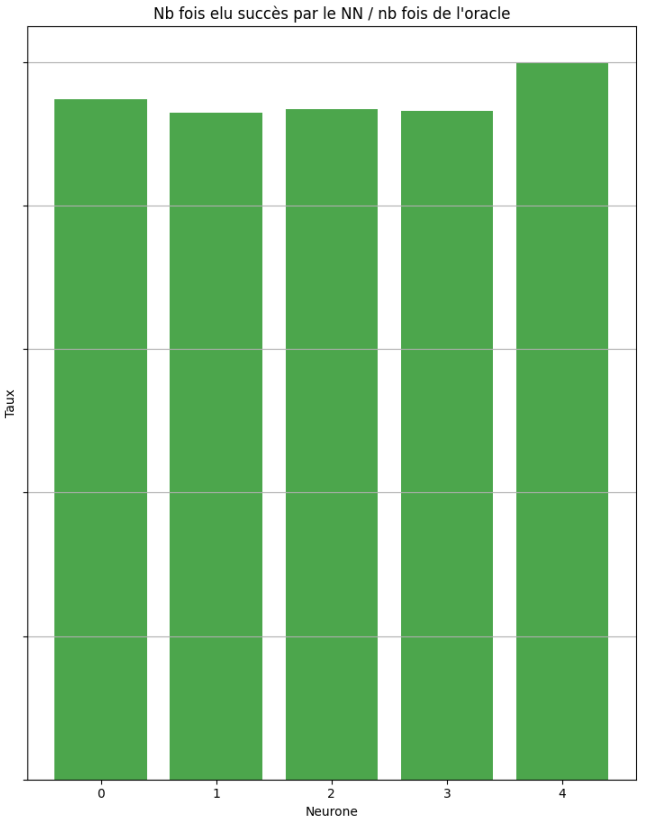


Taux succès lr = 0.0006



Choix des neurones lr = 0.0006

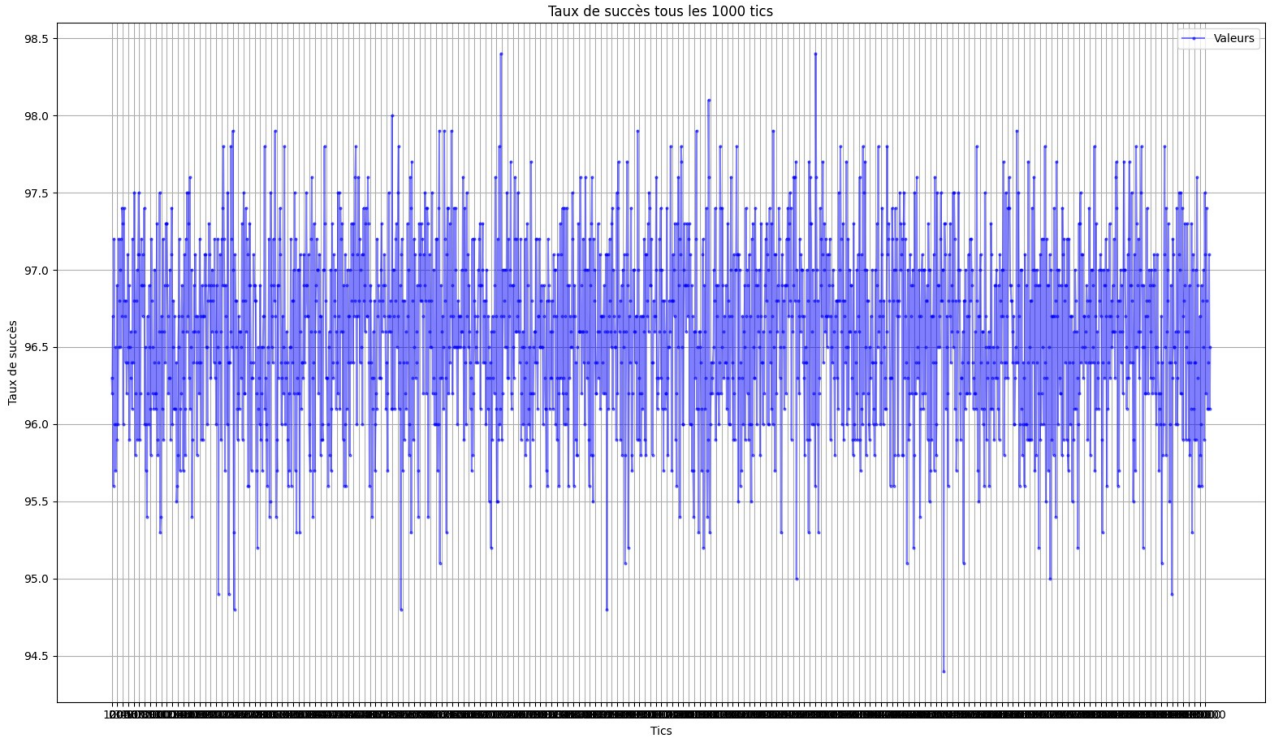




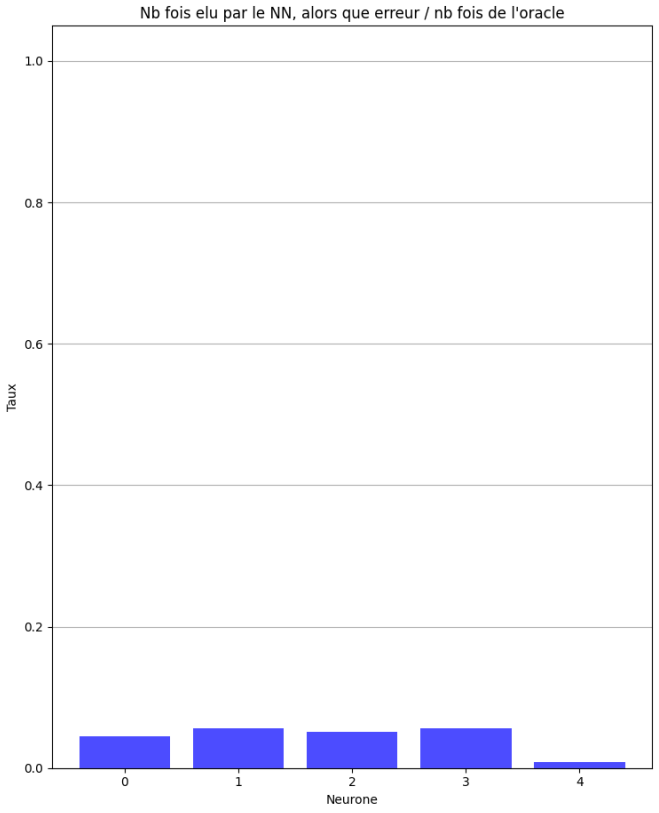
Gravité des erreurs lr = 0.0006

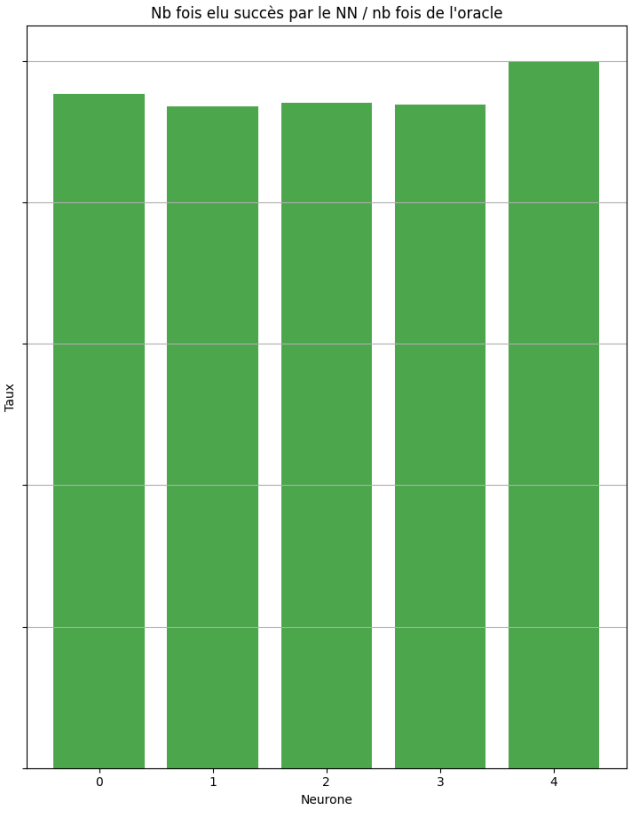


Taux succès lr = 0.0003

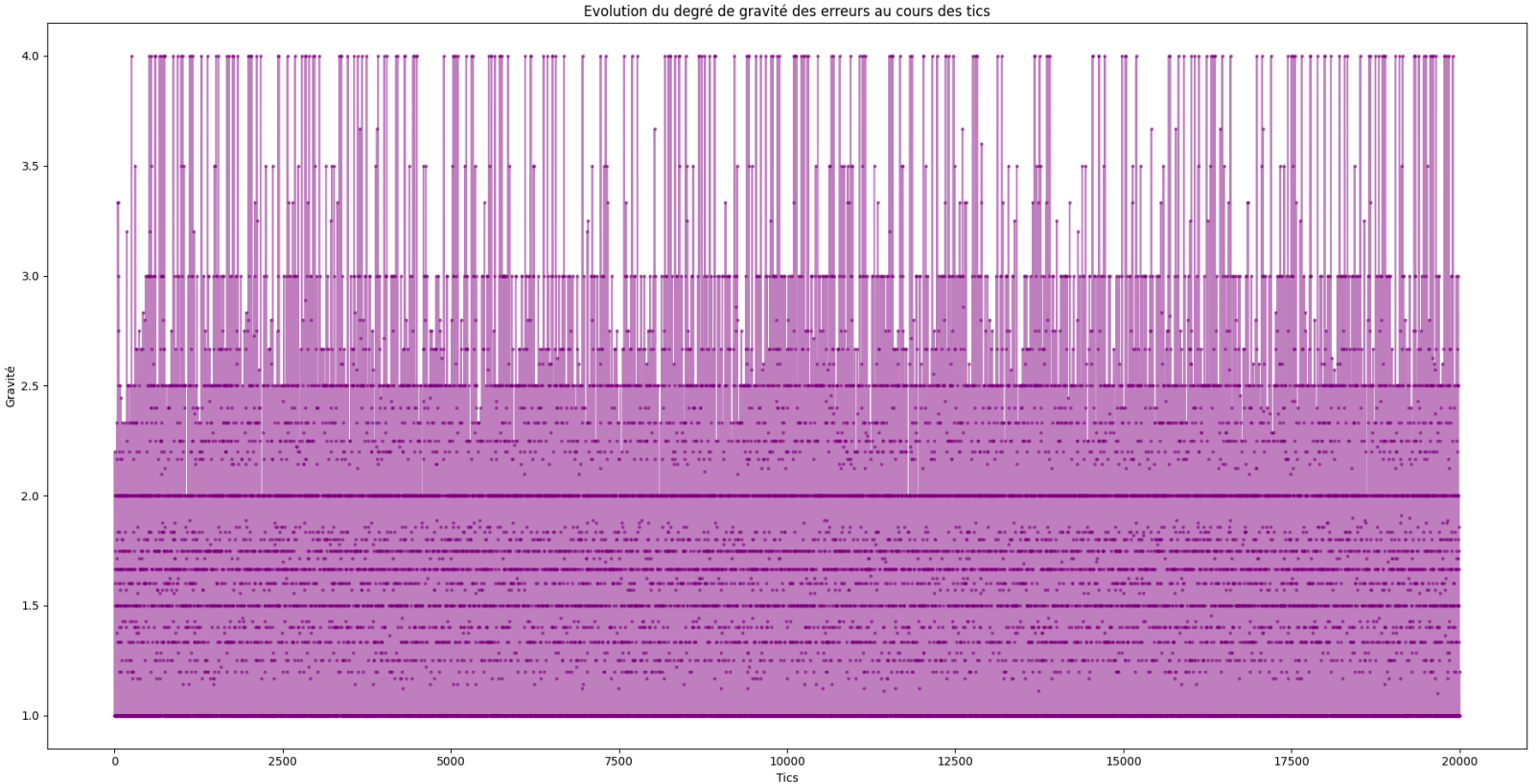


Choix des neurones lr = 0.0003

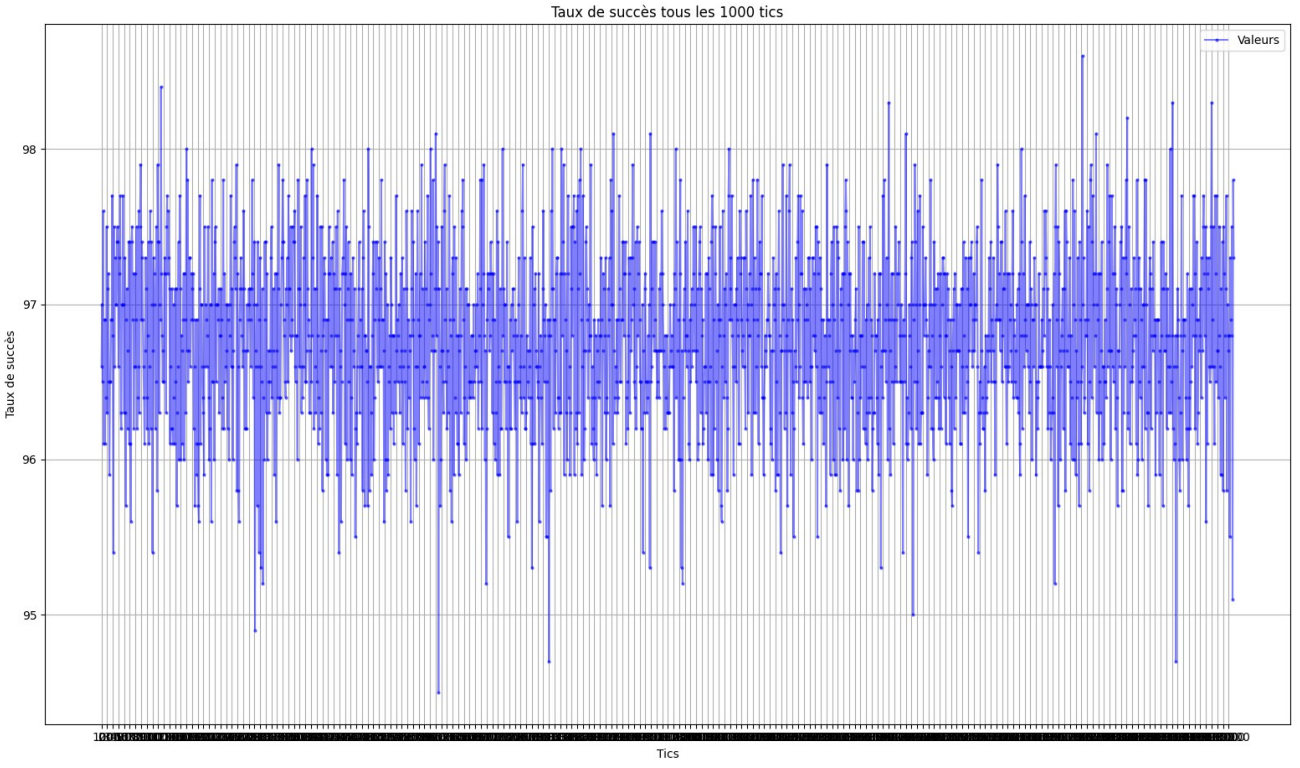




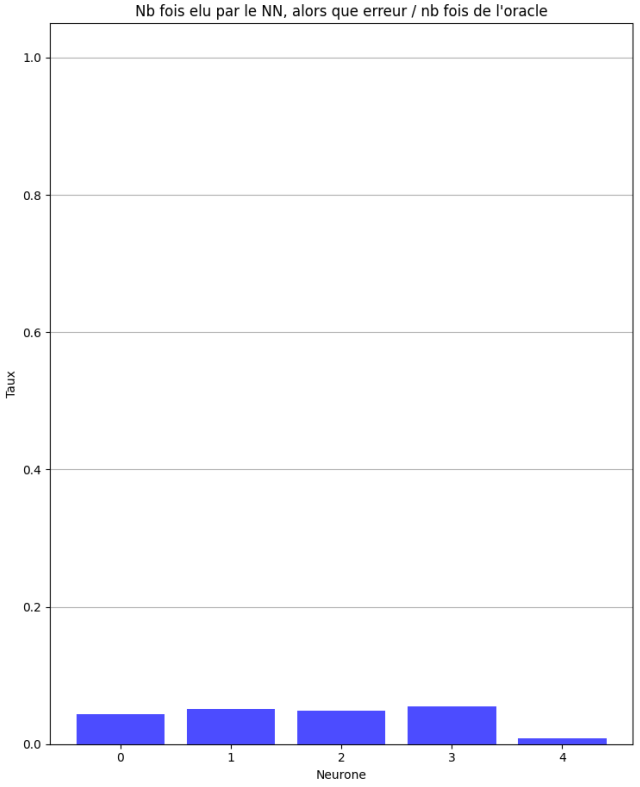
Gravité des erreurs lr = 0.0003

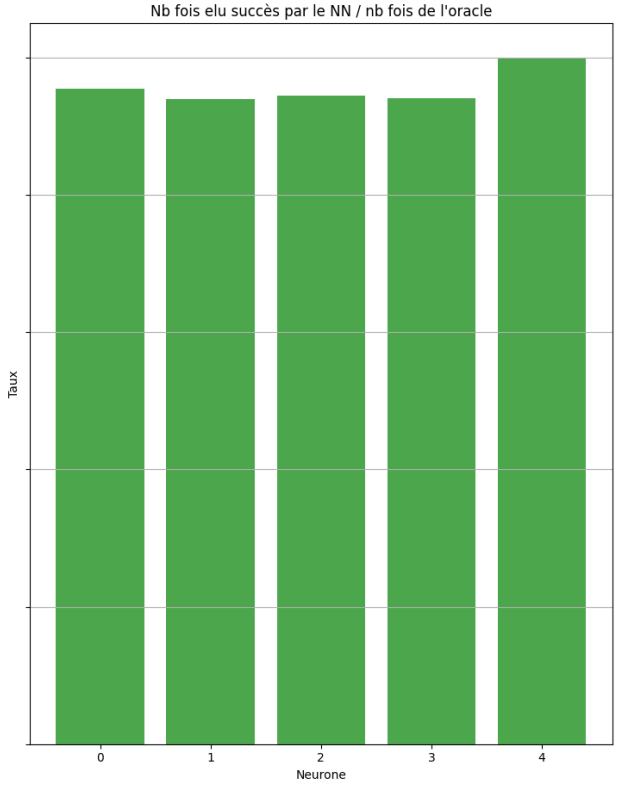


Taux succès lr = 0.00015

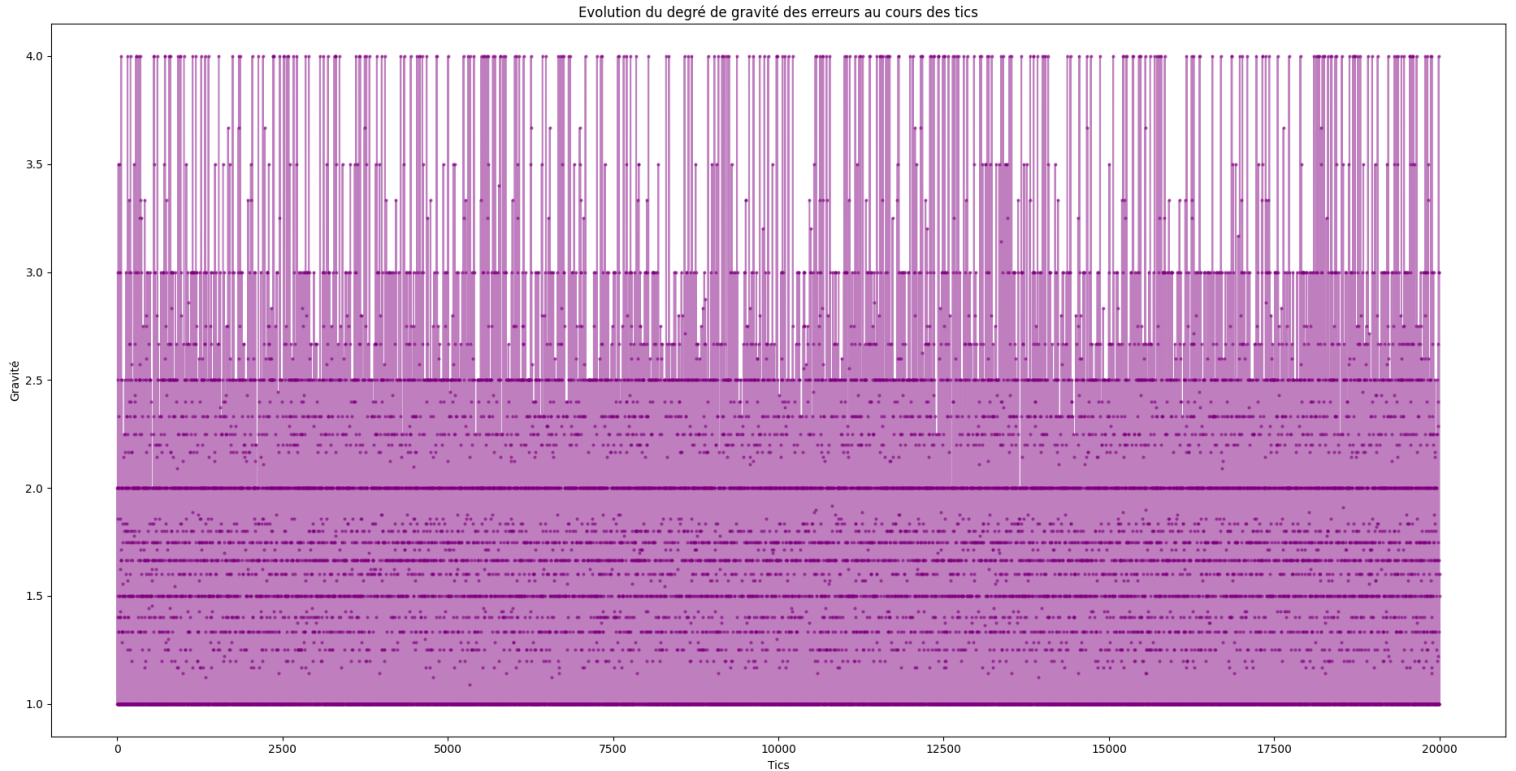


Choix des neurones lr = 0.00015





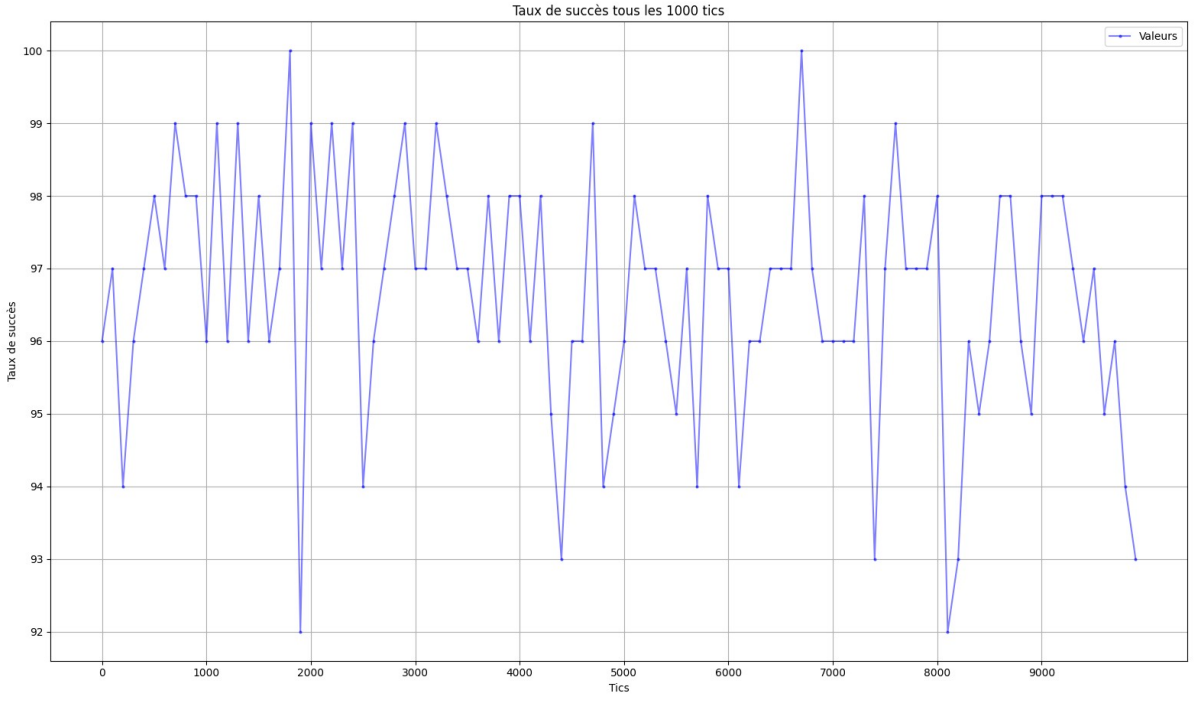
Gravité des erreurs lr = 0.00015



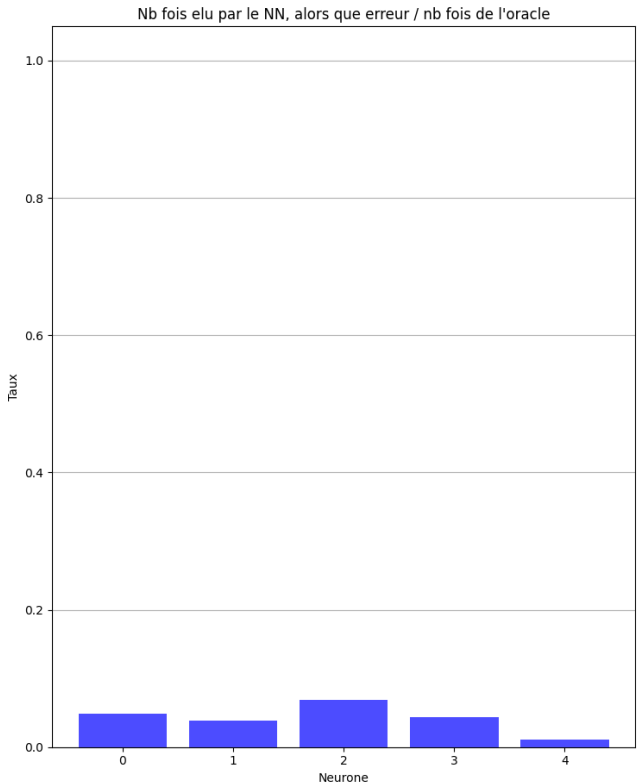
Évaluation et comparaison de notre réseau de neurones

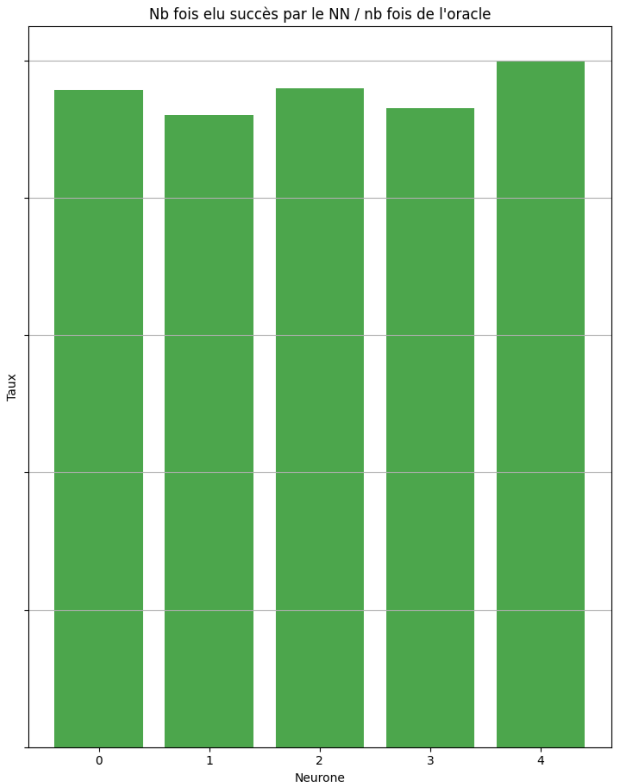
- Sur 10 000 tics
- Métriques :
  - Taux de succès
  - Neurones
  - Gravité erreurs
  - Taux de transmission des bits
  - Comparaison avec les quantités demandées

Taux de succès

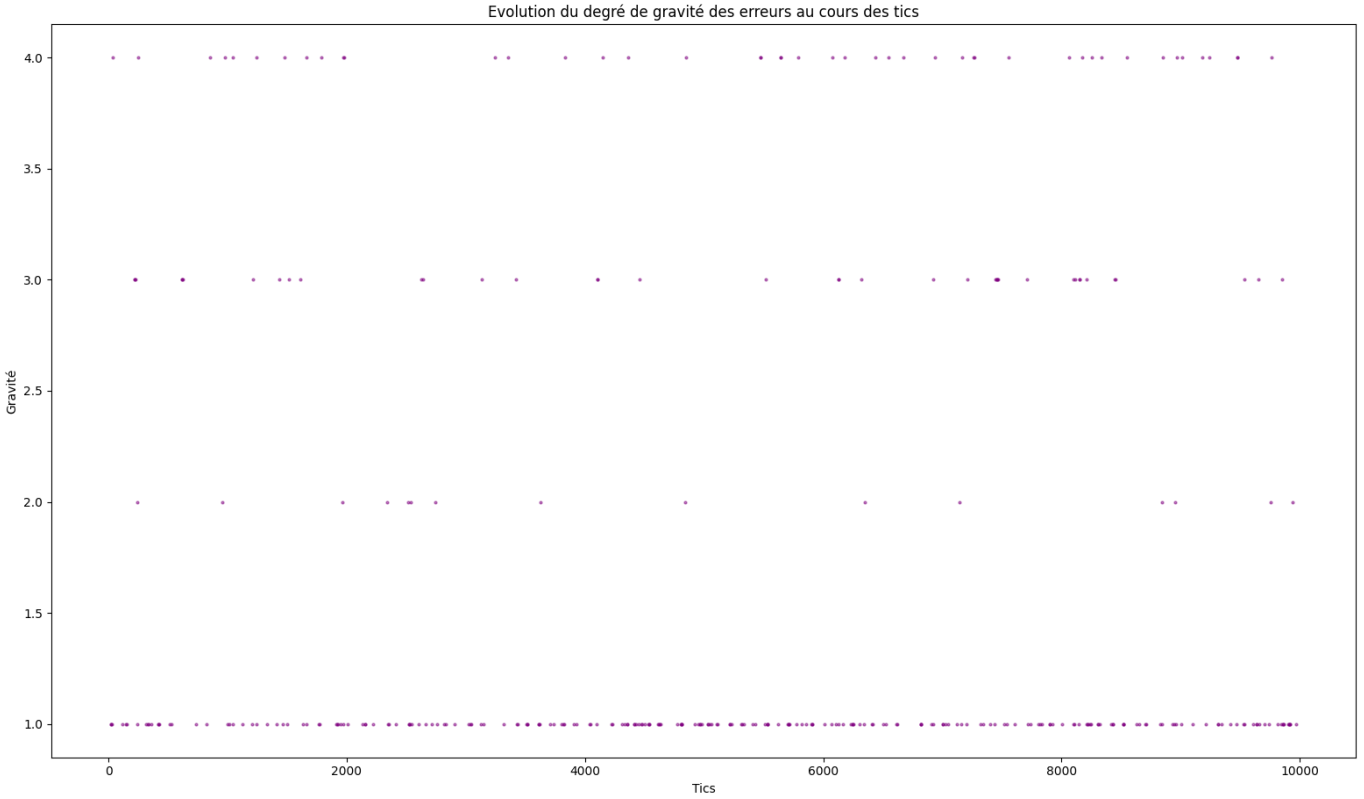


Neurones

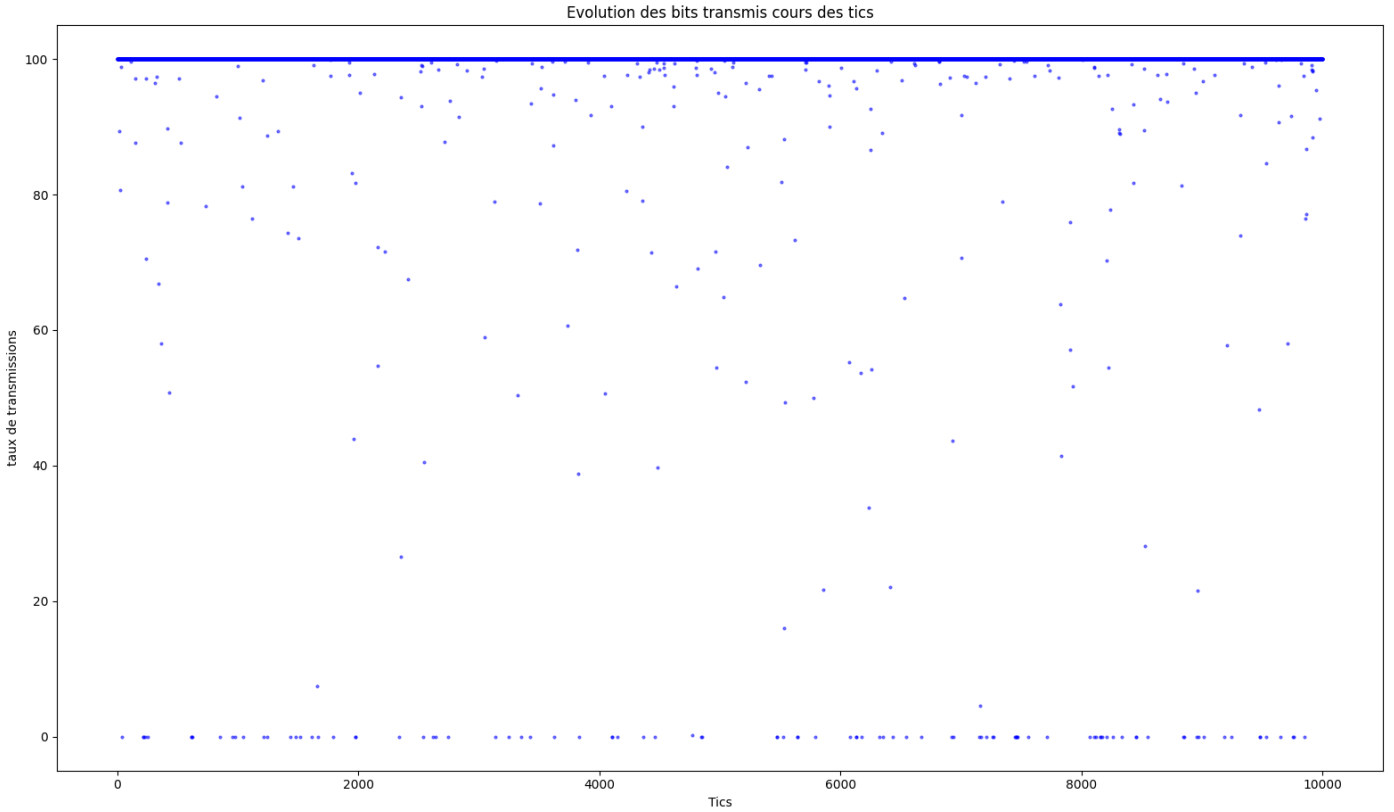

Nb fois elu par le NN, alors que erreur / nb fois de l'oracle
Taux
1.0
0.8
0.6
0.4
0.2
0.0
0
1
2
3
4
Neurone


Nb fois elu succès par le NN / nb fois de l'oracle
Taux
0
1
2
3
4
Neurone

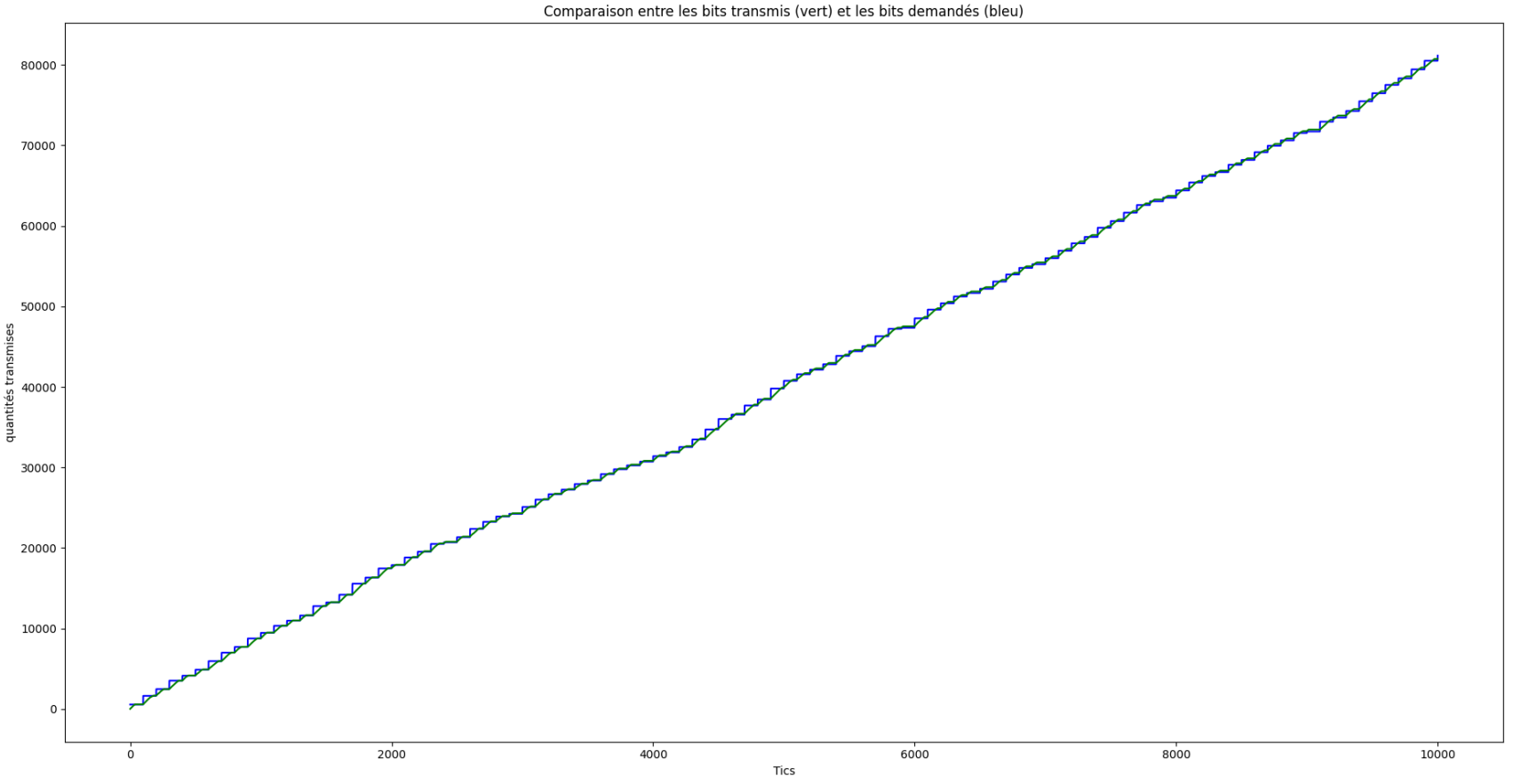
Gravité erreurs



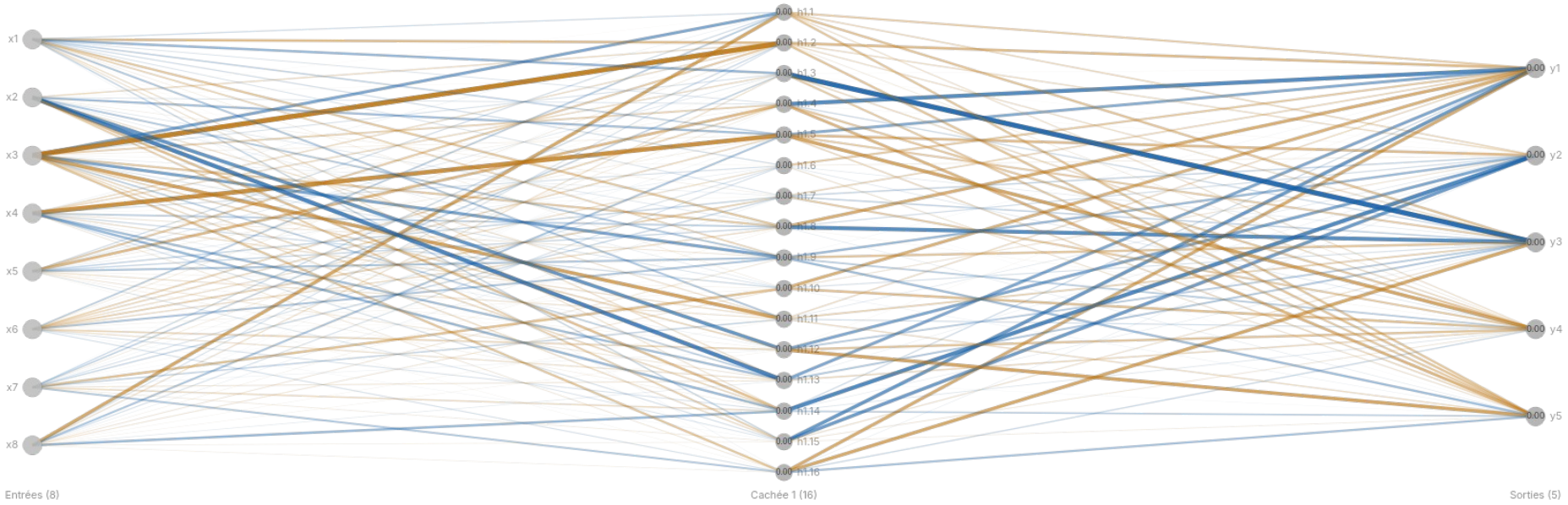
Taux transmission



Quantités transmises

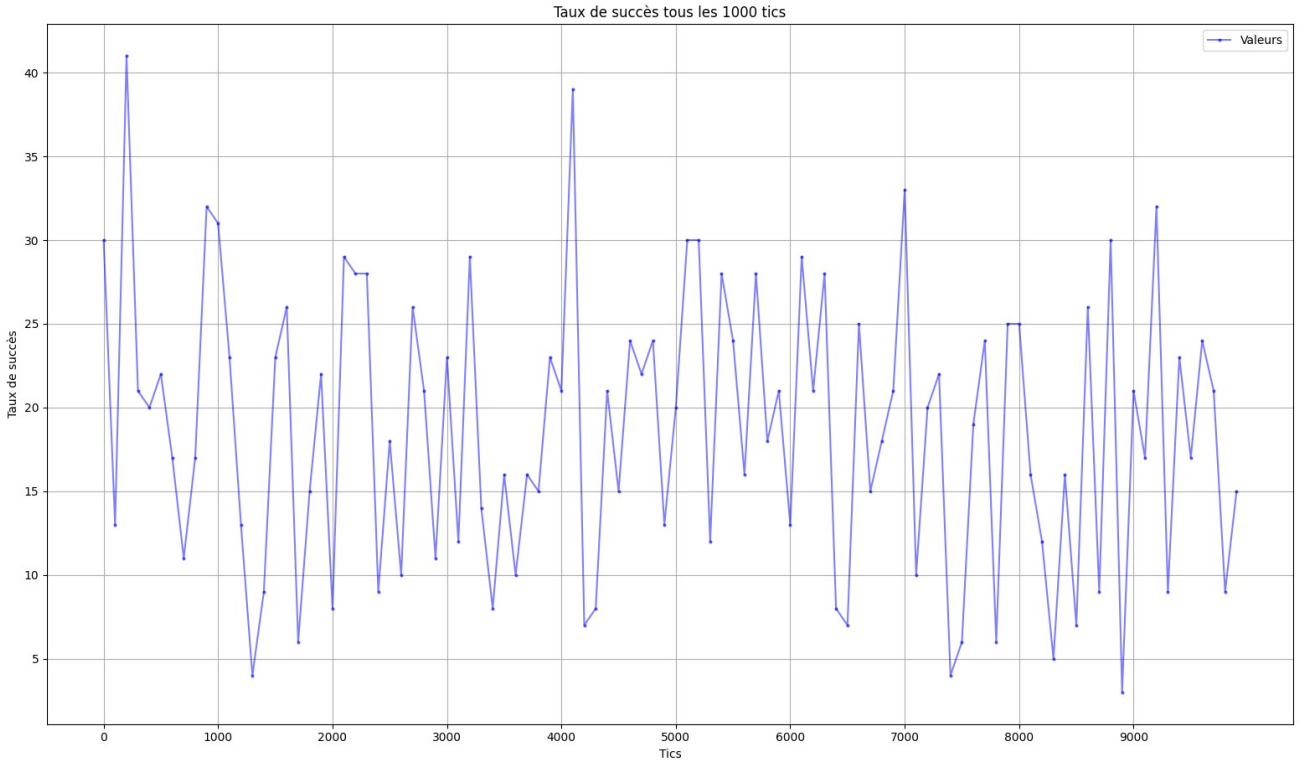


Sur un modèle non entrainé

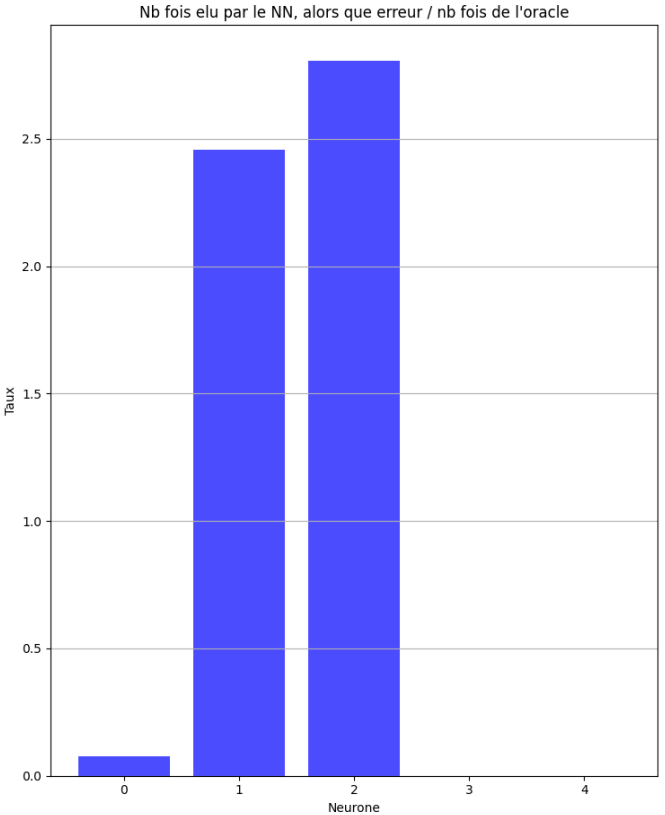


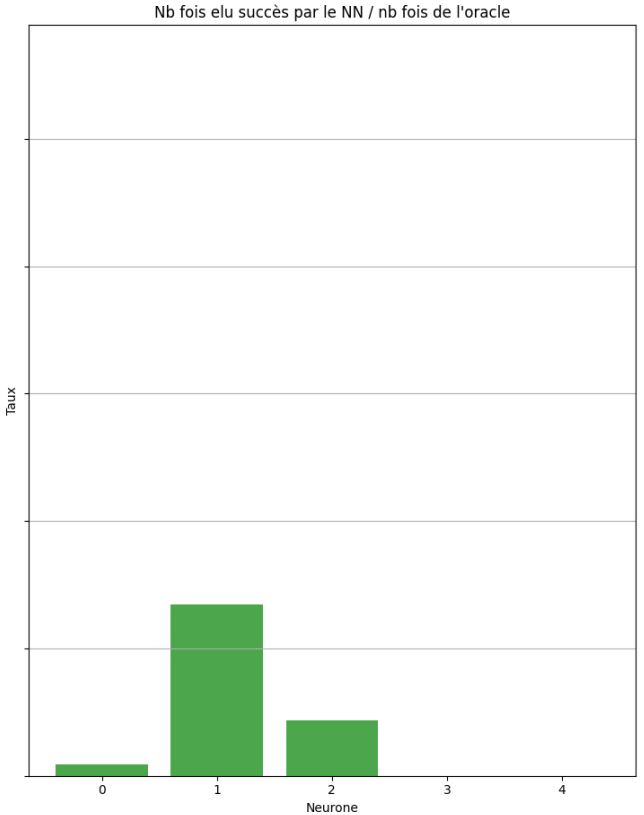
Poids positif Poids négatif Biais > 0 Biais < 0 Biais ≈ 0

Taux de succès

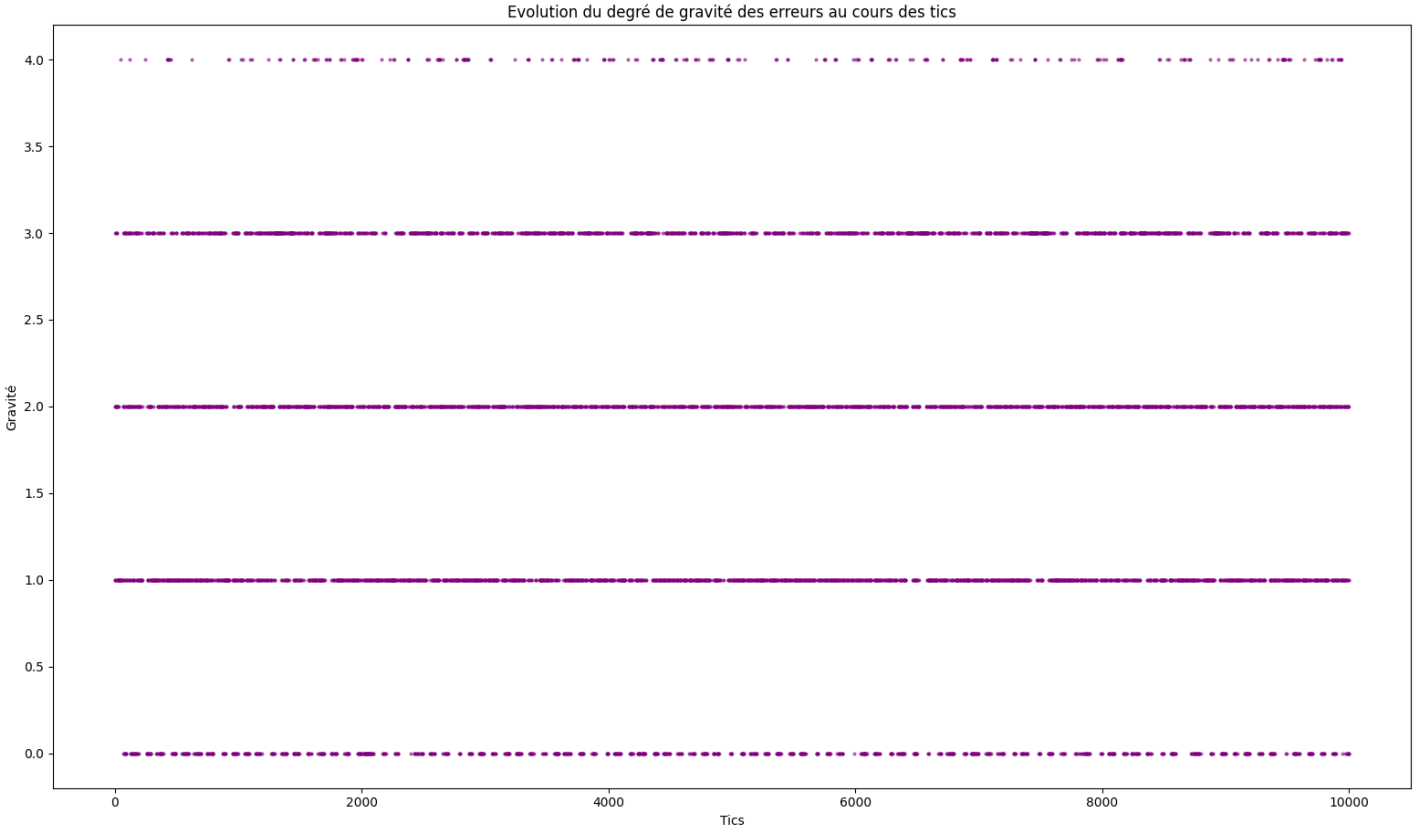


Neurones

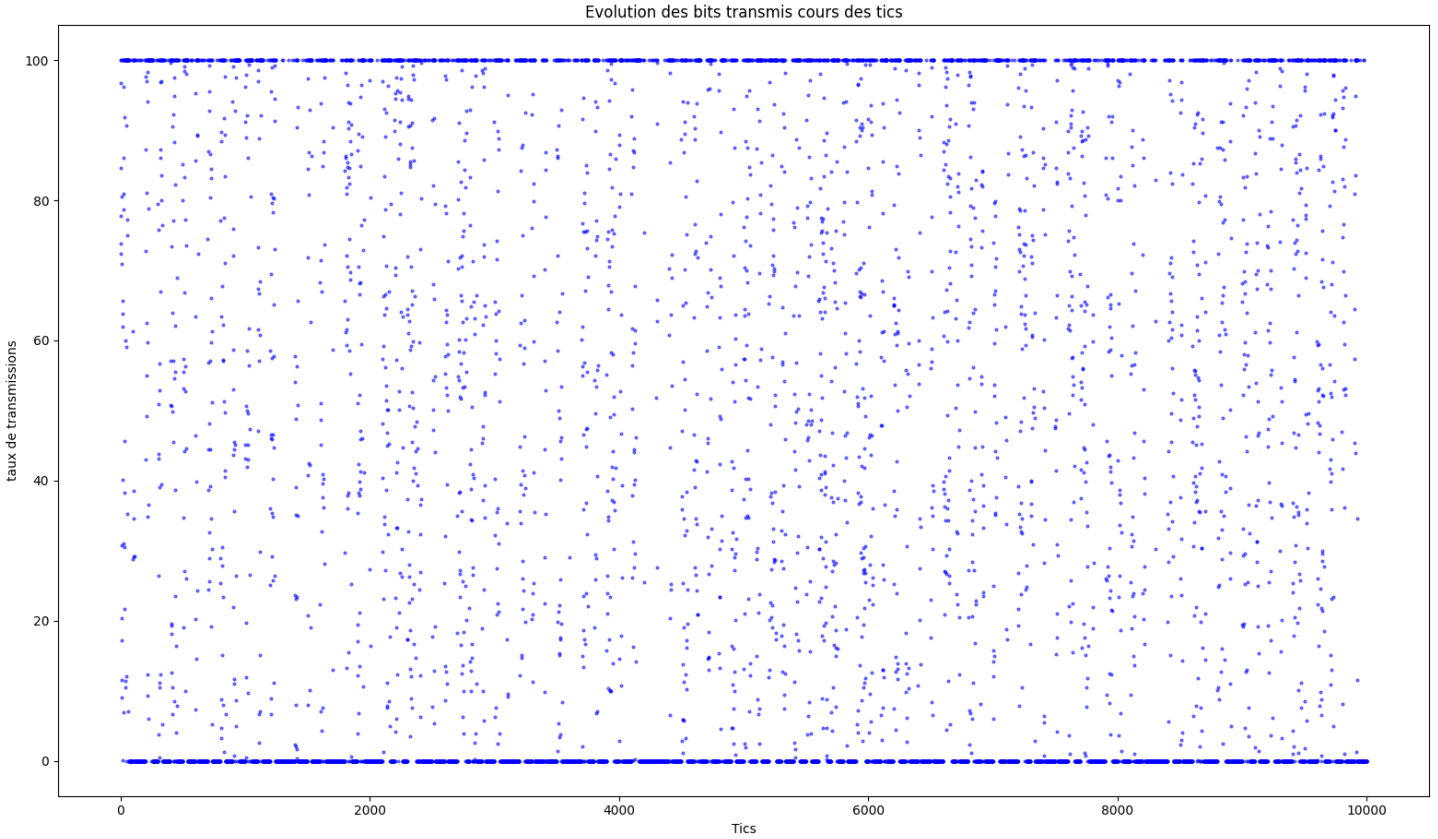

Nb fois elu par le NN, alors que erreur / nb fois de l'oracle
Taux
0.0
0.5
1.0
1.5
2.0
2.5
0
1
2
3
4
Neurone


Nb fois elu succès par le NN / nb fois de l'oracle
Taux
0
1
2
3
4
Neurone

Gravité erreurs



Taux transmission



Quantités transmises

